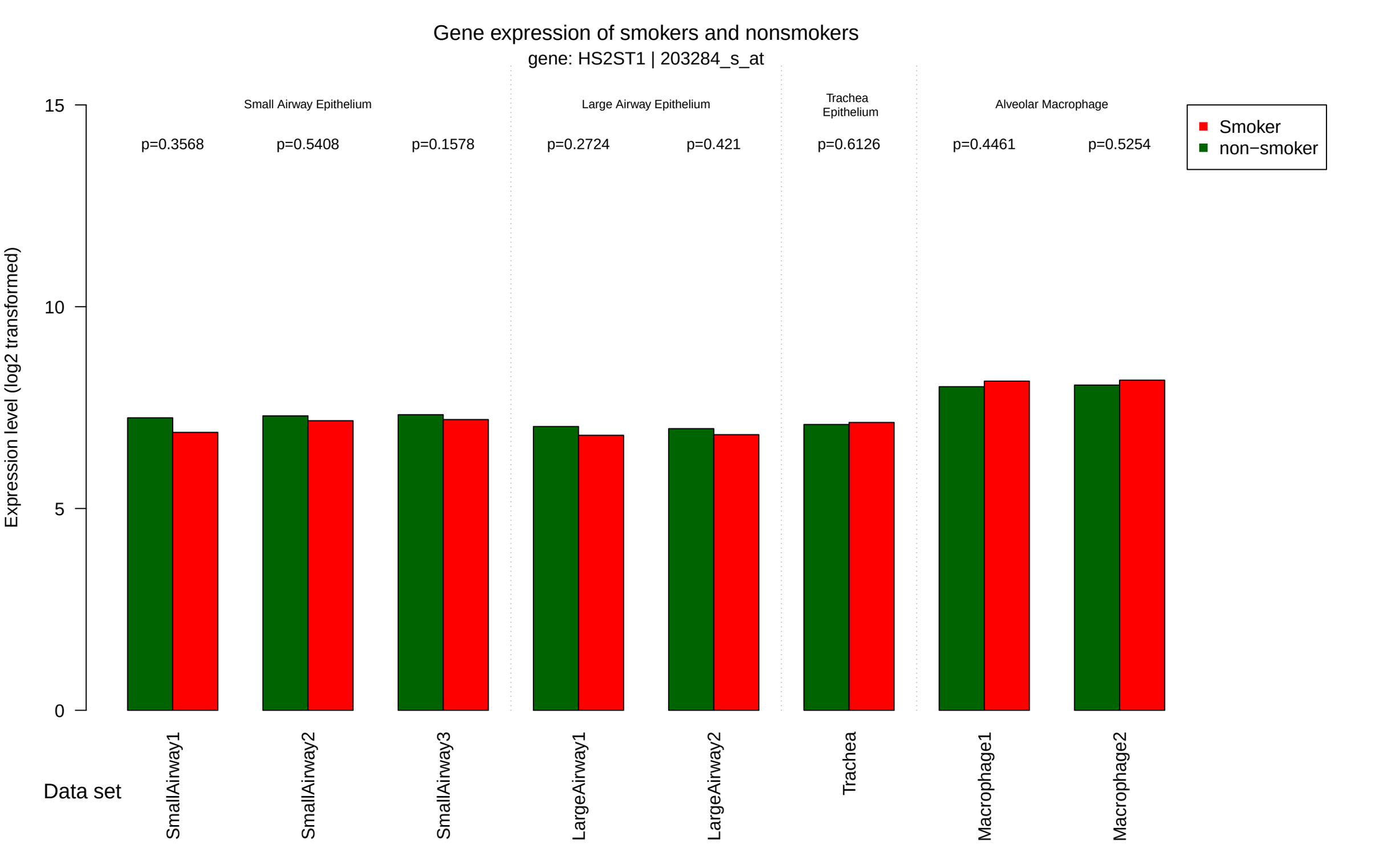
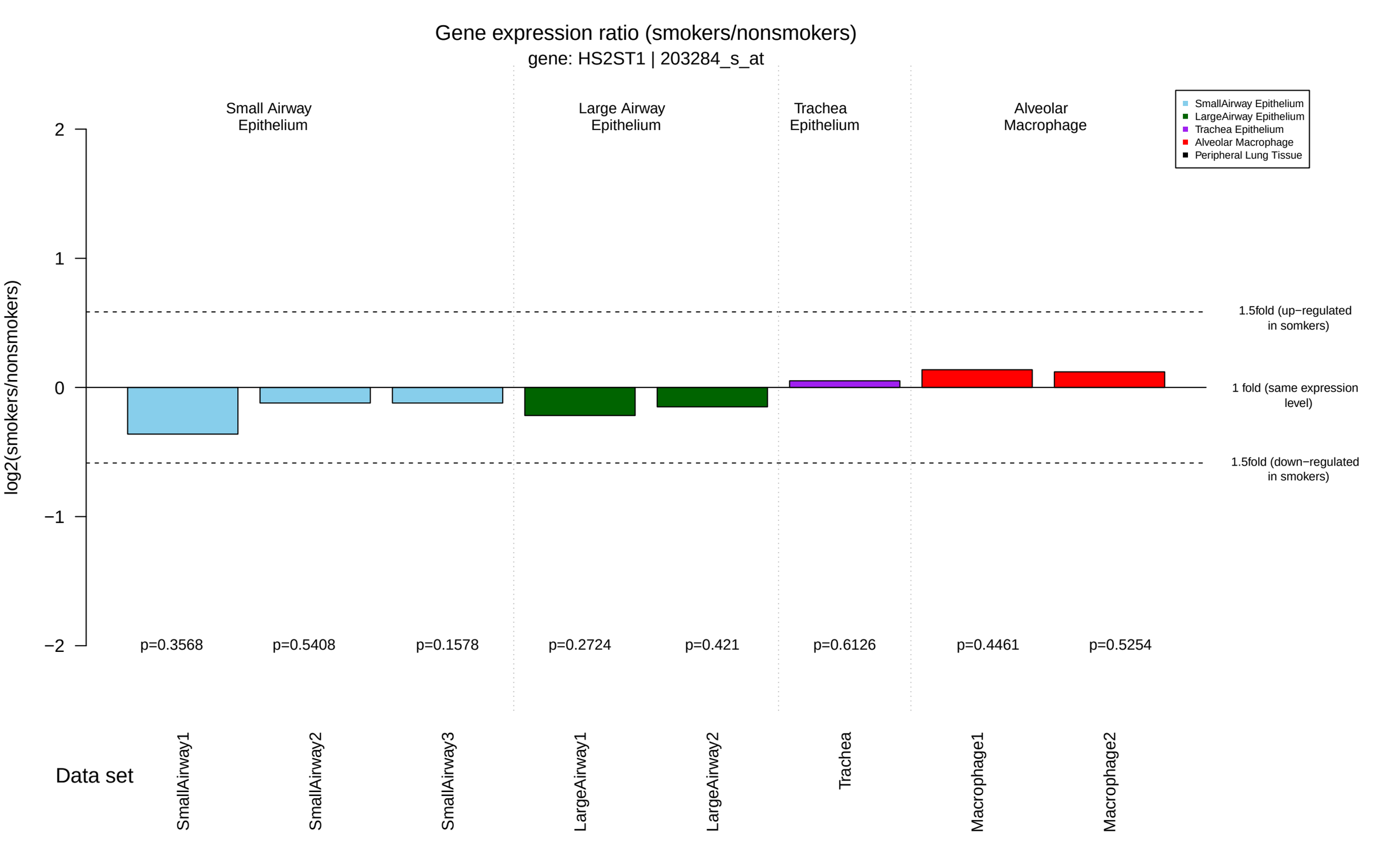
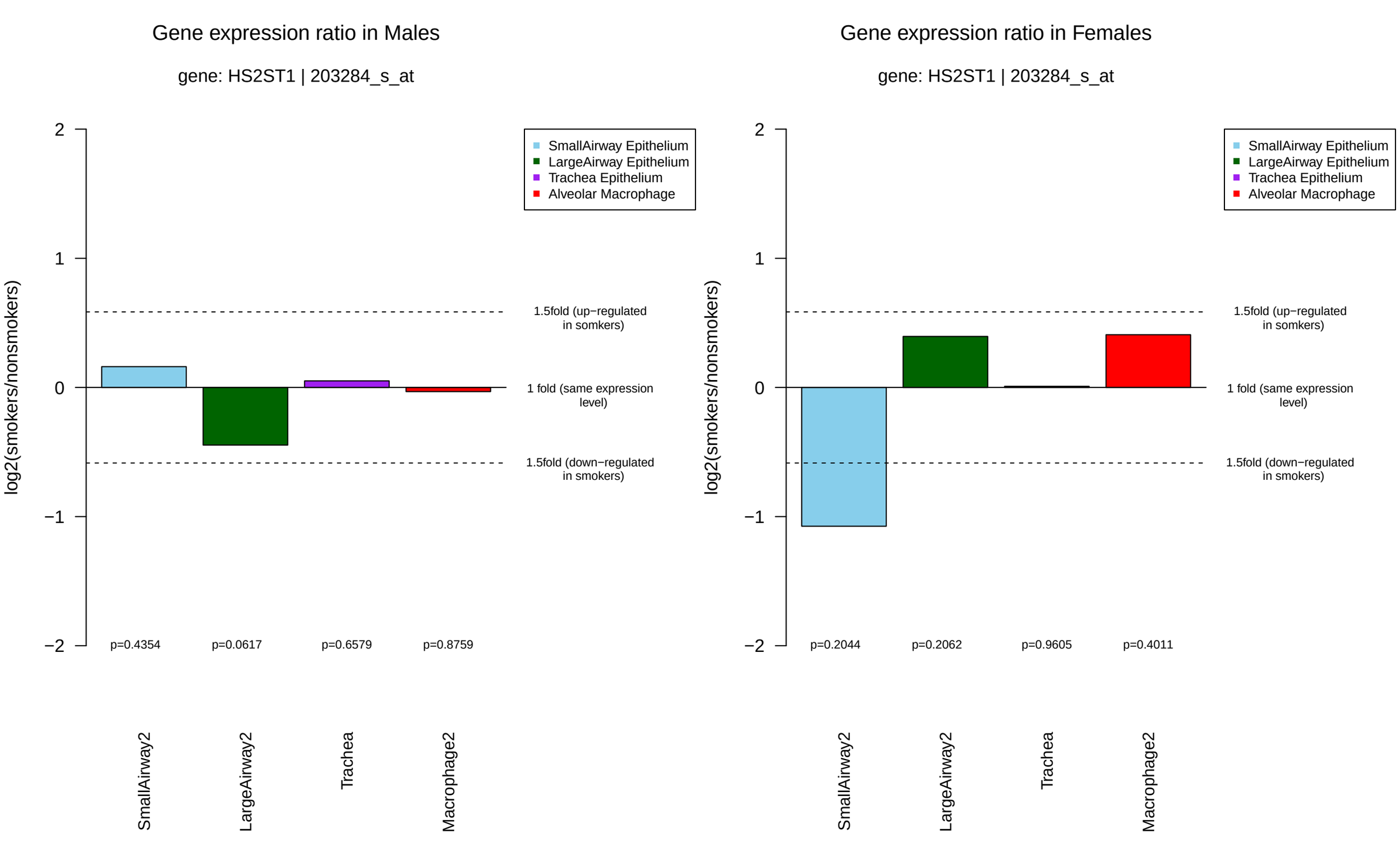
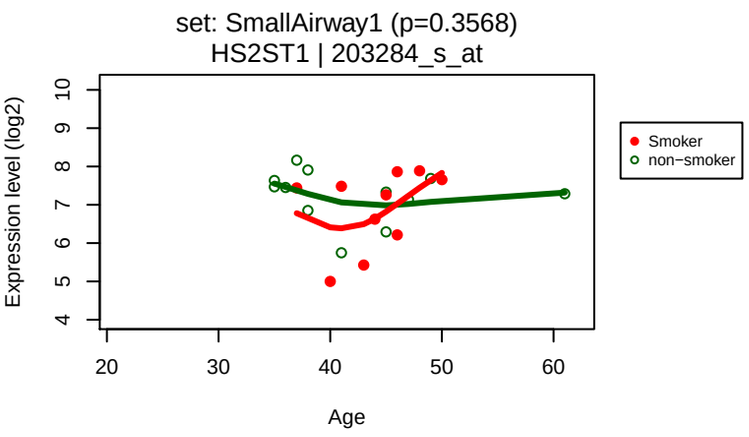
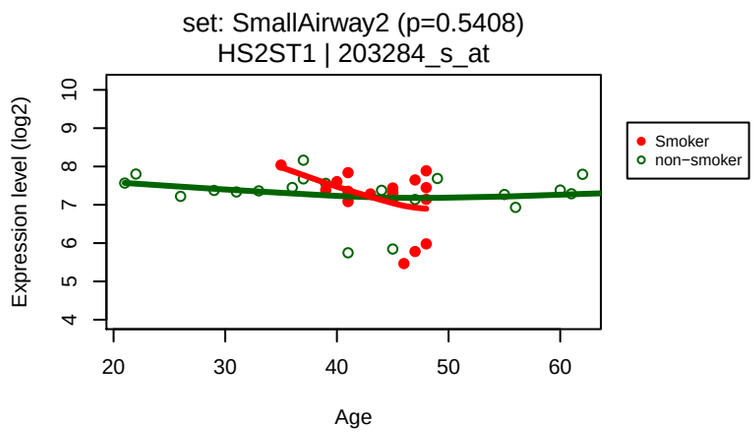
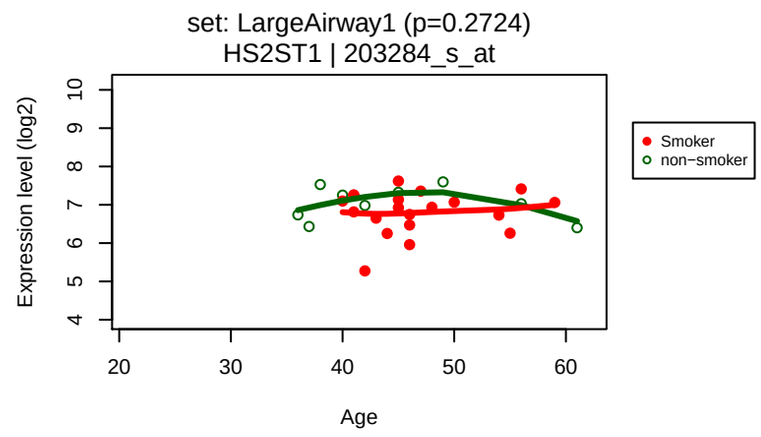

Gene expression of smokers and nonsmokers
gene: HS2ST1 | 203284_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3568
p=0.5408
p=0.1578
p=0.2724
p=0.421
p=0.6126
p=0.4461
p=0.5254
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

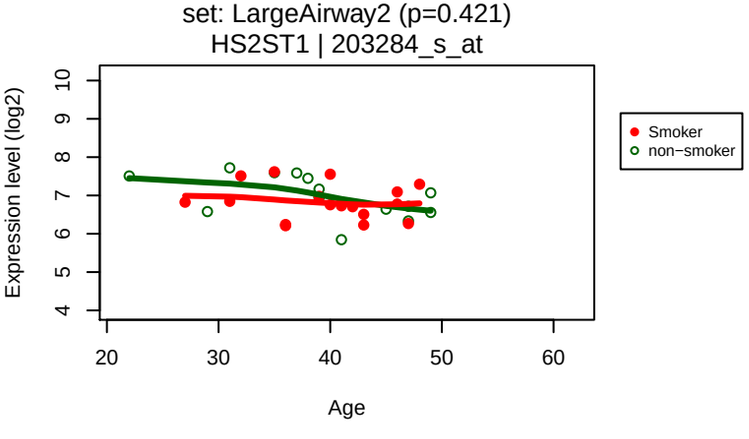

Gene expression ratio (smokers/nonsmokers)
gene: HS2ST1 | 203284_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3568
p=0.5408
p=0.1578
p=0.2724
p=0.421
p=0.6126
p=0.4461
p=0.5254
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

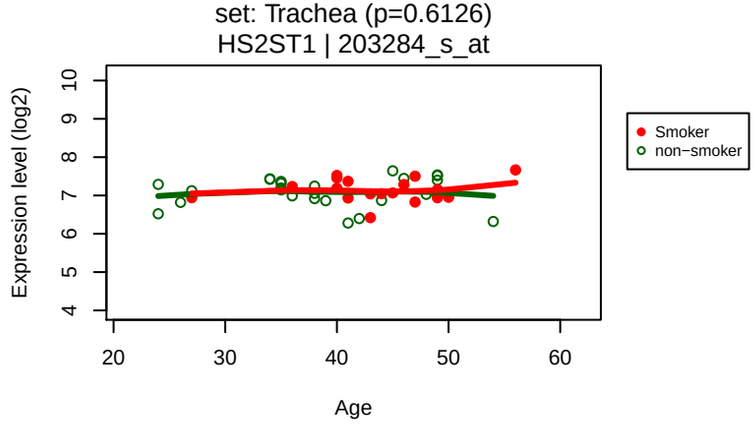

Gene expression ratio in Males
gene: HS2ST1 | 203284_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4354
p=0.0617
p=0.6579
p=0.8759
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HS2ST1 | 203284_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2044
p=0.2062
p=0.9605
p=0.4011
SmallAirway2
LargeAirway2
Trachea
Macrophage2

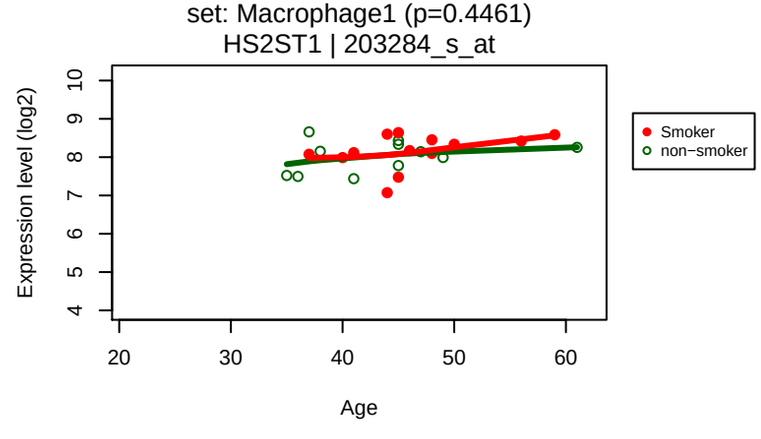

set: SmallAirway1 (p=0.3568)
HS2ST1 | 203284_s_at
Expression level (log2)
Age
Smoker
non−smoker

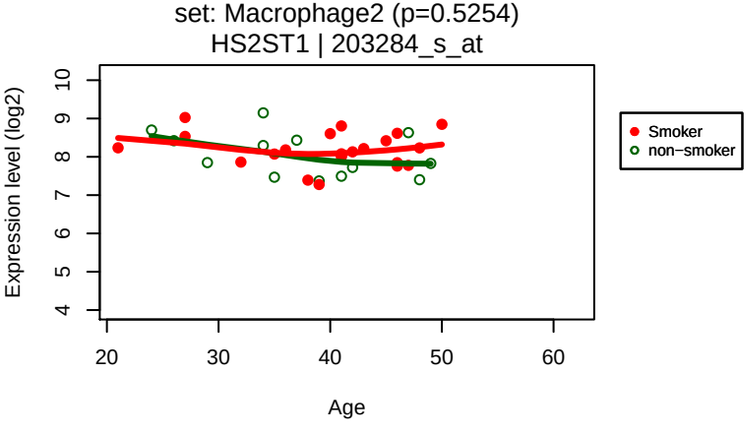

set: SmallAirway2 (p=0.5408)
HS2ST1 | 203284_s_at
Expression level (log2)
Age
Smoker
non−smoker

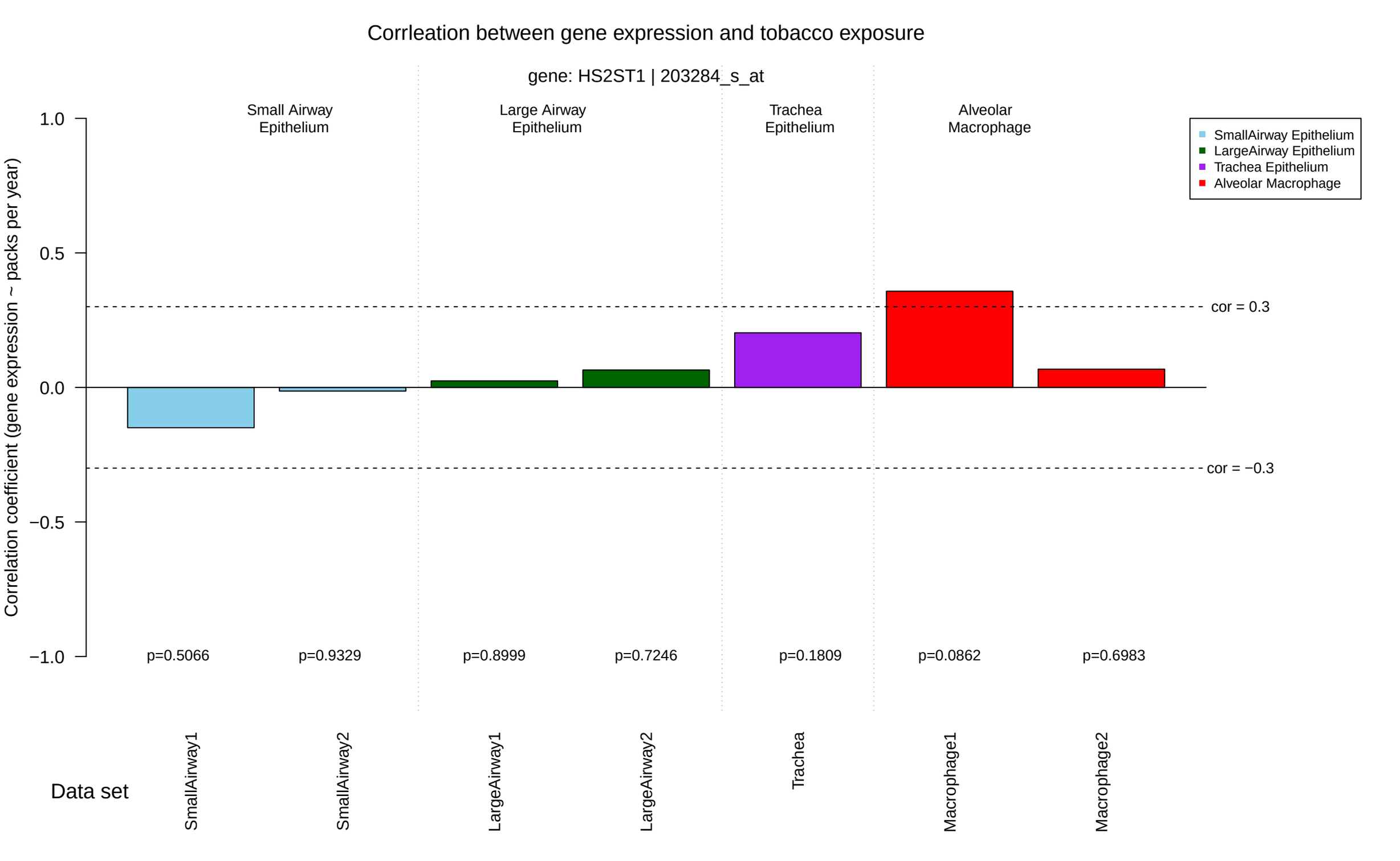

set: LargeAirway1 (p=0.2724)
HS2ST1 | 203284_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.421)
HS2ST1 | 203284_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6126)
HS2ST1 | 203284_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4461)
HS2ST1 | 203284_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5254)
HS2ST1 | 203284_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HS2ST1 | 203284_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5066
p=0.9329
p=0.8999
p=0.7246
p=0.1809
p=0.0862
p=0.6983
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2